

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tool

- Anti-seize, Part Number 3824879, or equivalent
- Aftertreatment Clean Care Kit, Part Number 5299193
- Diesel Exhaust Fluid (DEF) Extension Kit, Part Number 5299873
- Aftertreatment DEF Clean Care Kit, Part Number 5394236.

Additional Service Items

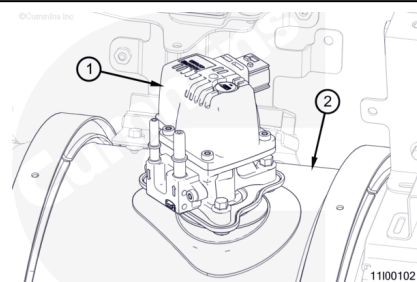
- Synthetic grease such as Super Lube®.

General Information

The aftertreatment DEF dosing valve (1) is used to spray DEF into the exhaust flow.

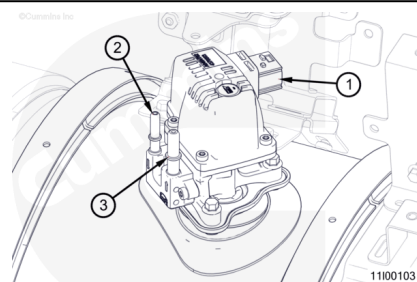
The aftertreatment DEF dosing valve is located on the aftertreatment decomposition tube (2).

When the aftertreatment DEF dosing valve is dosing DEF into the exhaust, there is a noticeable clicking sound. This is normal and should **not** be replaced if heard.



There are three primary connections on the aftertreatment DEF dosing valve.

1. 8 pin electrical connection
2. DEF supply line which connects the aftertreatment dosing valve to the aftertreatment DEF dosing unit
3. DEF return line which connects the aftertreatment dosing valve to the aftertreatment DEF tank.



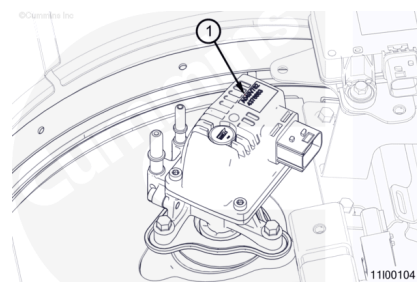
There are two different types of aftertreatment DEF dosing valves, 12V and 24V, depending on the application. Confirm that the correct part number (1) is used. The connections are

the same between the two different aftertreatment DEF dosing valves.

Reference the Aftertreatment Diesel Exhaust Fluid (DEF) Dosing Valve Identification Label Technical Service Bulletin, TSB150139, for part number verification.

For information on handling incorrect or contaminated DEF, Refer to Procedure 011-056 in Section 11.

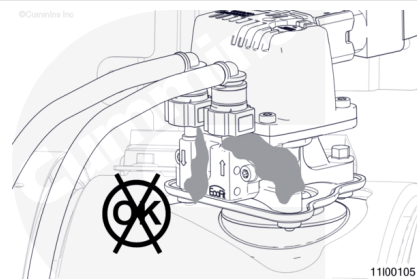
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Initial Check

⚠ CAUTION ⚠

Do not use the flow test portion of recommended Cummins® electronic service tool or equivalent Aftertreatment Diesel DEF Dosing Unit Override Test to check the system for leaks. This will spray DEF into the exhaust system at temperatures too low to evaporate, resulting in deposit formations in the exhaust system.



Note : DEF deposits could possibly be left over from a previous DEF spill or repair. Verify active leaks before replacing any components.

Inspect the area around the DEF line connection location at the aftertreatment DEF dosing valve.

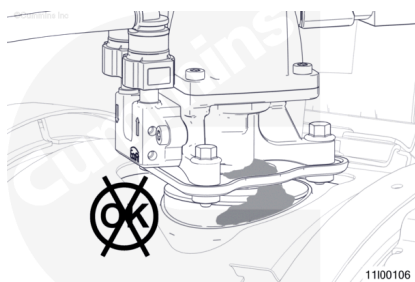
Inspect for signs of leaks and/or white deposits.

Due to deposit buildups possibly masking the source of the leak, it can be necessary to remove the deposits and clean the area with warm water.

Once the source of the DEF leaks and/or deposits are identified, see the Clean and Inspect for Reuse section of this procedure.

Inspect the body and around the base of the aftertreatment DEF dosing valve. Check for signs of exhaust leaks and white deposits.

If leaks and/or deposits are identified, see the Clean and Inspect for Reuse section of this procedure.



Preparatory Steps

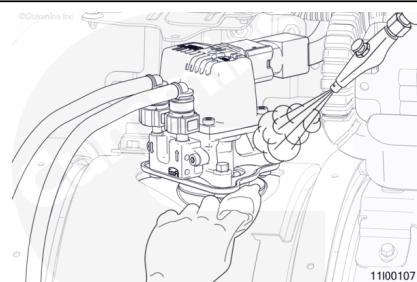
⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Allow the DEF to recirculate and depressurize after engine shut down and prior to DEF system servicing. This can take up to 15 minutes.
- Disconnect the batteries. See equipment manufacturer service information.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



⚠ WARNING ⚠

Diesel exhaust fluid (DEF) contains urea. Do not get the substance in your eyes. In case of contact, immediately flush eyes with large amounts of water for a minimum of 15 minutes. Do not swallow. In the event the diesel exhaust fluid (DEF) is ingested, contact a physician immediately. Reference the Materials Safety Data Sheet (MSDS) for additional information.

Use compressed air to remove any debris from around the aftertreatment DEF dosing valve and connections.

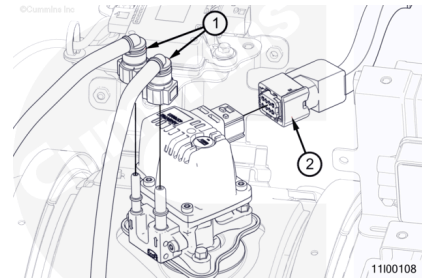
Remove any debris trapped in the gaps between the aftertreatment DEF dosing valve and the aftertreatment decomposition tube so that debris does **not** fall into the

aftertreatment decomposition tube upon removal of the aftertreatment DEF dosing valve.

If there are any white deposits on the DEF line connection, remove them by wiping with a clean shop towel soaked in warm water.

⚠ CAUTION ⚠

Care should be taken when handling and/or disconnecting the DEF line from the aftertreatment DEF dosing valve. The DEF supply line connector of the aftertreatment DEF dosing valve can be easily damaged.



There is **always** a chance of residual pressure being present. Open fittings slowly to allow any pressure to bleed off before removing any connections.

Disconnect the connections at the aftertreatment DEF dosing valve in the following order:

Place a small plastic container under the aftertreatment DEF dosing valve to catch any residual DEF prior to disconnecting the DEF supply line.

1. Disconnect the DEF line. See equipment manufacturer service information.
2. Pull the yellow locking tab on the connector to disconnect the aftertreatment wiring harness from the aftertreatment DEF dosing valve.

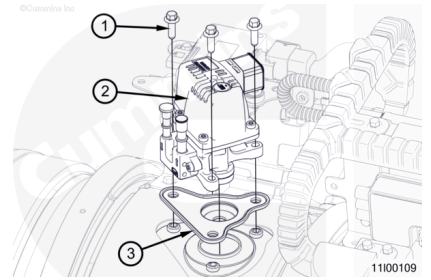
Plug the aftertreatment DEF lines with plugs from Aftertreatment Clean Care Kit, Part Number 5299193, to reduce the possibility of debris entering the system.

After the electrical connection is disconnected, cover the connection on the aftertreatment DEF dosing valve with electrical tape to prevent DEF from getting into the electrical connection.

Remove

⚠ CAUTION ⚠

Do not set the aftertreatment DEF dosing valve down on the spray area of the aftertreatment DEF dosing valve. Damage can occur.



Remove the capscrews (1).

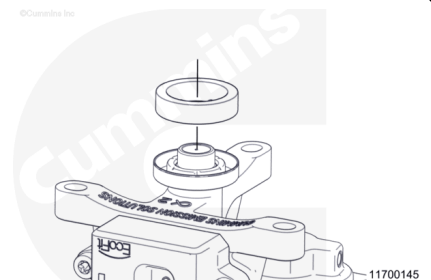
Remove aftertreatment DEF dosing valve (2).

Remove and discard the thermal isolator (3).

Note : A small amount of DEF deposits is normal if observed on the aftertreatment DEF dosing valve face and around the ring seal of the aftertreatment DEF dosing valve.

Use a pick to carefully remove the aftertreatment DEF dosing valve nozzle ring seal.

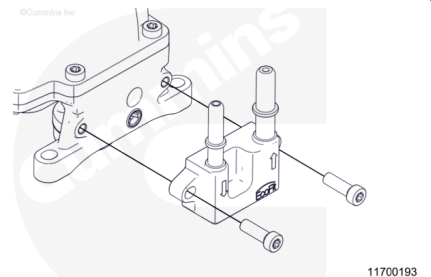
Discard the ring seal.



Disassemble

⚠ CAUTION ⚠

A small amount of dirt and debris can be very harmful to the DEF dosing system. Dirt or other foreign objects must not get into the Aftertreatment DEF dosing valve.

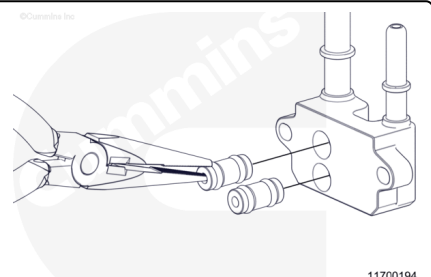


Use a T-25 Torx™ socket drive to remove the capscrews from the aftertreatment DEF dosing valve connector manifold.

Discard the capscrews.

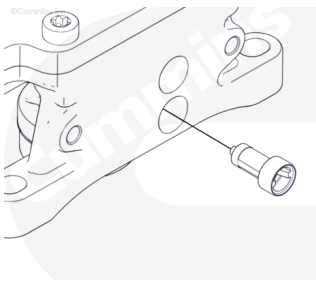
Remove the connector cylinders with needle nose pliers.

Discard the connector cylinders.



Use needle nose pliers to remove the filter from the aftertreatment DEF dosing valve connector manifold.

Discard the filter.



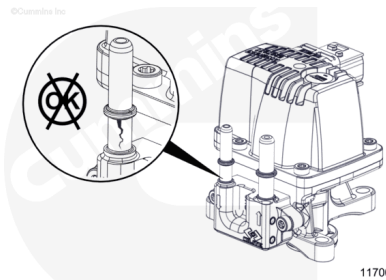
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Clean and Inspect for Reuse

If leaks or deposits are found at the aftertreatment DEF dosing valve joints between the valve and the valve assembly body during the Initial Check section of this procedure, replace the aftertreatment DEF dosing valve.

If leaks are suspected from the aftertreatment DEF dosing valve nozzle tip, perform the Aftertreatment Diesel Exhaust Fluid System Leak Test with the aftertreatment DEF dosing valve removed. Refer to Procedure 011-080 in Section 11.
(/qs3/pubsys2/xml/en/procedures/377/377-011-080.html)

Count the number of drops from the aftertreatment DEF dosing valve within 30 seconds.



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Aftertreatment DEF Dosing Valve Leak Test

Max Drops	Seconds
3	30

If leaks or deposits are found at the DEF line connection during the Initial Check section of this procedure, inspect the DEF line connection port for cracks or pitting.

If damage or corrosion is found, also inspect the DEF line connection. See equipment manufacturer service information.

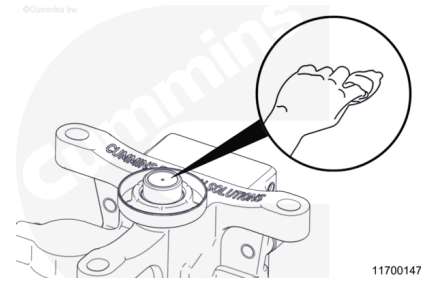
Inspect the electrical connection and pins for damage and corrosion. Refer to Procedure 019-361 in Section 19.
(/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)

If damage or corrosion is found, inspect the electrical connector and pins on the vehicle harness. See equipment manufacturer service information.

Replace the aftertreatment DEF dosing valve if damage is found.

⚠ CAUTION ⚠

Use only warm water to clean the tip of the aftertreatment DEF dosing valve. The use of a wire wheel or brass brush or any other abrasive medium will cause permanent damage to the aftertreatment DEF dosing valve.



Use a clean shop cloth soaked in warm water to dissolve and wipe DEF deposits away on the aftertreatment DEF dosing valve tip.

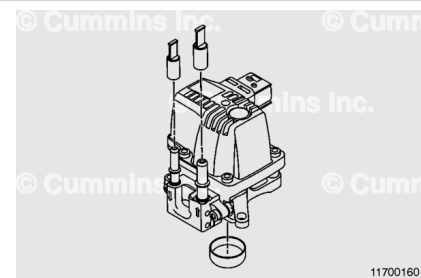
Inspect the aftertreatment DEF dosing valve tip for cracks or other damage. Replace if damage is found.

If leaks or deposits are found at the base of the aftertreatment DEF dosing valve during the Initial Check section of this procedure, inspect the following:

- Check the bottom of the aftertreatment DEF dosing valve around the ring seal sealing surface for signs of heavy corrosion, pitting, and/or surface damage. Replace the aftertreatment DEF dosing valve if damage is found.
- Check the aftertreatment DEF dosing valve mounting surface on the decomposition chamber.

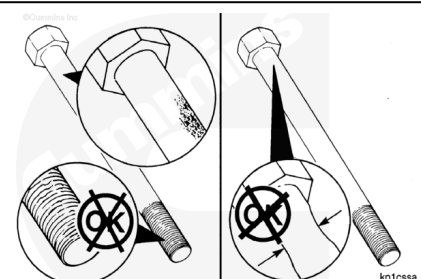
If replacing the aftertreatment DEF dosing valve, make sure to transfer the DEF Clean Care Kit, Part Number 5394236, protective caps from the replacement aftertreatment DEF dosing valve to the old aftertreatment DEF dosing valve prior to putting it in the core return box.

Make sure to fill out any requested information (engine serial number (ESN), hours, and vehicle identification number (VIN)).

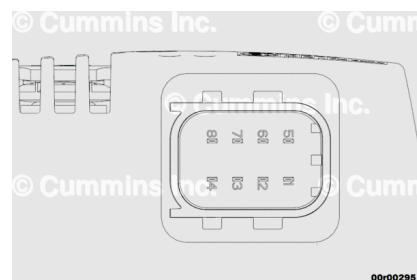


Inspect the capscrews for damaged threads and corroded surfaces.

Replace if damage is found.



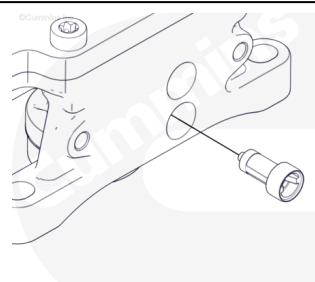
Measure the resistance between the aftertreatment DEF dosing valve SIGNAL pin (2) and aftertreatment DEF dosing valve RETURN pin (1) with aftertreatment DEF dosing valve disconnected from the original equipment manufacturer (OEM) harness. Replace the aftertreatment DEF dosing valve if measured resistance does **not** meet the following specification:



System Voltage (VDC)	Resistance (ohms)	
	Minimum	Maximum
12	4.5	9.5
24	12	22

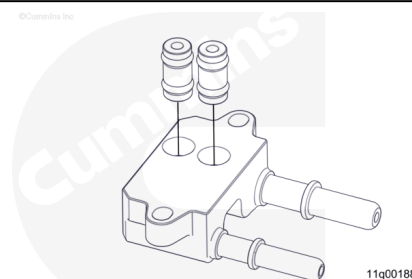
Assemble

Install a new aftertreatment DEF dosing valve filter in the bottom hole of the valve assembly. The filter will **only** fit in the bottom hole.



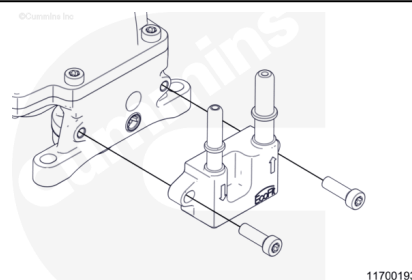
Apply synthetic grease onto the o-rings.

Install new connector cylinders into the connector manifold. Make sure to press the cylinders until they contact the bottom of the connector manifold. The cylinders are identical and can go in either hole.



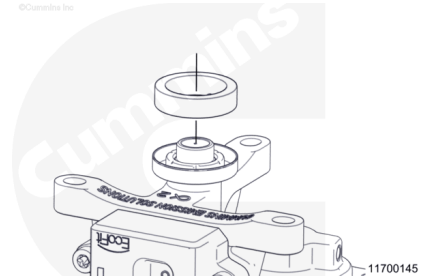
Install the connector manifold onto the aftertreatment DEF dosing valve assembly using two new Torx™ capscrews.

Torque Value: 8 n•m [71 in-lb]



Install

Install a new aftertreatment DEF dosing valve nozzle ring seal onto the dosing nozzle.



Remove any protective caps.

Make sure the aftertreatment DEF dosing valve spray port is free of debris.

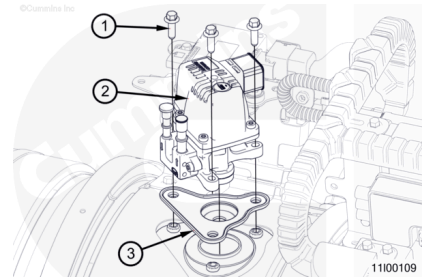
Apply a coat of high temperature anti-seize compound, Part Number 3824879, to the threads of the aftertreatment DEF dosing valve mounting capscrews.

Install the thermal isolator (3).

Install the aftertreatment DEF dosing valve (2).

Install and tighten the capscrews (1).

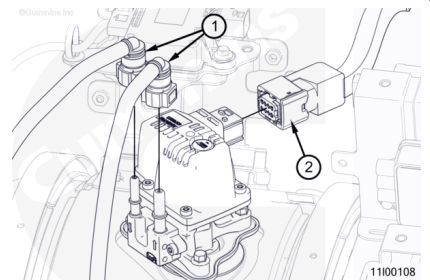
Torque Value: 10 n·m [89 in-lb]



Finishing Steps

⚠ CAUTION ⚠

Do not use the flow test portion of recommended Cummins® electronic service tool or equivalent Aftertreatment Diesel DEF Dosing Unit Override Test to check the system for leaks. This will spray DEF into the exhaust system at temperatures too low to evaporate, resulting in deposit formations in the exhaust system.



Remove any protective caps and/or covers from the electrical, coolant, and DEF ports immediately prior to making the connection.

Connect the electrical connector to the aftertreatment DEF dosing valve.

Lubricate the DEF line fittings with a drop of synthetic grease such as Super Lube®.

Connect the DEF lines (1). See equipment manufacturer service information.

Note : DEF forms a white residue when allowed to dry. This residue can prevent proper sealing of the fittings when the DEF line is installed. Before making the DEF line connection:

- Dissolve DEF residue off the DEF line fittings with distilled water.
- Inspect the DEF line fitting o-rings for cuts or damage. Replace the DEF lines if there is any damage to the o-rings.
- Lubricate the DEF line fittings with a drop of synthetic grease.

Once all connections are made, give each a slight pull to make sure they are securely installed.

** WARNING **

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Connect the batteries. See equipment manufacturer service information.
- Operate the engine. Check for leaks.